

Math Thermo
class #11
02/24/2026

Recap

For N non-interacting particles, with exactly r accessible (discrete) energy levels, the Gibbs entropy,

$$S_G(\vec{n}) = -k_B \sum_{j=1}^r n_j \log\left(\frac{n_j}{N}\right),$$

is maximized by the Maxwell-Boltzmann distribution,

$$n_j^{\text{eq}} = \exp\left(\frac{-U_j}{k_B T}\right) \cdot \frac{N}{Z},$$

where Z is the partition function given by

$$Z = Z(T) = \sum_{i=1}^r \exp\left(\frac{-U_i}{k_B T}\right).$$

Recall that the mass and energy constraints hold

$$N = \sum_{j=1}^r n_j^{\text{eq}}, \quad U = \sum_{j=1}^r n_j^{\text{eq}} U_j,$$

where the energy levels $U_j, j=1, \dots, r$, are fixed. All of the Thermodynamic variables are given in terms of Z :

$$F(T, N) = -N k_B T \ln(Z),$$

$$\tilde{U}(T, N) = N k_B T^2 \frac{\partial}{\partial T} [\log(Z)],$$

$$\tilde{S}_{\text{eq}}(T, N) = N k_B \log(Z) + N k_B T \frac{\partial}{\partial T} [\log(Z)],$$

$$\mu = k_B T \log(Z).$$

Continuous Energy Levels

Next we consider the case that the energy levels are continuous.

Again, we will only consider the case of N non-interacting particles. The energy of each particle is assumed to be a function of the form

$$(11.1) \quad u(\vec{x}, \vec{v}) = \frac{1}{2} m |\vec{v}|^2 + \Psi(\vec{x}),$$

where $\vec{x} \in \Omega$ is the position of the particle in the domain Ω and $\vec{v} \in \mathbb{R}^3$ is its velocity vector.

The energy of any one particle is independent of the position and velocity of the other particles in the system, hence non-interacting.

m is the mass of the particle; $\vec{v}$ and $\vec{x}$ are its velocity and position vectors, resp.

Ψ is a potential energy, which depends upon only the position of the particles.

The term $\frac{1}{2} m |\vec{v}|^2$ is called the kinetic energy.

The expression (11.1) is an idealization because it neglects energetic interactions between particles.

If N is large, it is too hard to track particles individually. We use a density instead.

Definition (11.1): let $\Omega \subseteq \mathbb{R}^3$ be a simply - connected domain with volume

$$V = \int_{\Omega} d\vec{x}.$$

The number density of particles is a function

$$f: \Omega \times \mathbb{R}^3 \rightarrow [0, \infty)$$

with the property

$$(11.2) \quad \int_{\Omega} \int_{\mathbb{R}^3} f(\vec{x}, \vec{v}) d\vec{v} d\vec{x} = N > 0$$

where N is the total number of particles in Ω .
The function

$$p: \Omega \times \mathbb{R}^3 \rightarrow [0, \infty),$$

defined as

$$p(\vec{x}, \vec{v}) := \frac{f(\vec{x}, \vec{v})}{N},$$

is called the particle probability function. of course

$$(11.3) \quad \int_{\Omega} \int_{\mathbb{R}^3} p(\vec{x}, \vec{v}) d\vec{v} d\vec{x} = 1.$$

We assume that there is a fixed and finite amount of energy in the system

$$(11.4) \quad U = \iint_{\Omega \times \mathbb{R}^3} f(\vec{x}, \vec{v}) u(\vec{x}, \vec{v}) d\vec{v} d\vec{x}$$

The set $Q = \Omega \times \mathbb{R}^3$ is called phase space. It is of course in $\mathbb{R}^6$!

Suppose $A \subseteq Q$. The number of particles in A is

$$N \geq N_A := \iint_A f(\vec{x}, \vec{v}) d\vec{v} d\vec{x}.$$

The energy of the particles in that set is denoted

$$U \geq U_A := \iint_A f(\vec{x}, \vec{v}) u(\vec{x}, \vec{v}) d\vec{v} d\vec{x}.$$

Definition (11.2): The average energy (per particle) is defined as

$$\bar{U} = \iint_Q f(\vec{x}, \vec{v}) u(\vec{x}, \vec{v}) d\vec{v} d\vec{x}.$$

Of course, it must be that

$$(11.5) \quad U = N \bar{U}.$$

Note

$$[p] = \frac{S^3}{m^6} = \frac{1}{[\vec{v}]^3 [\vec{x}]^3} = \frac{1}{[\text{Vol Phase Space}]}.$$

Definition (11.3): The Gibbs entropy for a system of N particles in phase space $Q = \Omega \times \mathbb{R}^3$ with particle density $f: Q \rightarrow [0, \infty)$ is defined as

$$(11.6) \quad S_G[f] := -k_B \iint_Q f(\vec{x}, \vec{v}) \log\left(\frac{f(\vec{x}, \vec{v})}{P_0 N}\right) d\vec{v} d\vec{x},$$

where $P_0 > 0$ is a normalization factor with the property that

$$\left[\frac{f(\vec{x}, \vec{v})}{P_0 N} \right] = \left[\frac{p(\vec{x}, \vec{v})}{P_0} \right] = 1 \Leftrightarrow [P_0] = \frac{1}{[\vec{v}]^3 [\vec{x}]^3}$$

The average entropy (per particles) is defined as

$$\bar{S}_G[f] = \frac{S_G[f]}{N}.$$

Thus,

$$(11.7) \quad \bar{S}_G[f] = -k_B \iint_Q p(\vec{x}, \vec{v}) \log\left(\frac{p(\vec{x}, \vec{v})}{P_0}\right) d\vec{v} d\vec{x}.$$

Remark: The arguments for a transcendental function, like $\log$, must be unitless.

As in the discrete (quantized) energy case the equilibrium state is the one that maximizes

the Gibbs entropy.

Theorem (11.1): The equilibrium particle density for non-interacting particles with constraints

$$(11.8) \quad N = \iint_Q f(\vec{x}, \vec{v}) d\vec{v} d\vec{x}$$

and

$$(11.9) \quad U = \iint_Q f(\vec{x}, \vec{v}) u(\vec{x}, \vec{v}) d\vec{v} d\vec{x}$$

is given by the Maxwell-Boltzmann distribution

$$(11.10) \quad f(\vec{x}, \vec{v}) = f^{\text{eq}}(\vec{x}, \vec{v}) = P_0 \frac{N}{Z} \exp\left(\frac{-\beta u(\vec{x}, \vec{v})}{k_B}\right),$$

where Z is the partition function

$$(11.11) \quad Z = P_0 \iint_Q \exp\left(\frac{-\beta u(\vec{x}, \vec{v})}{k_B}\right) d\vec{v} d\vec{x}$$

and β is a Lagrange multiplier that can be uniquely determined as a function of U and N .

Proof: let us minimize the functional

$$\begin{aligned} J[f] = & k_B \iint_Q f(\vec{q}) \log\left(\frac{f(\vec{q})}{P_0 N}\right) d\vec{q} \\ & + \alpha \left\{ \iint_Q f(\vec{q}) d\vec{q} - N \right\} + \beta \left\{ \iint_Q f(\vec{q}) u(\vec{q}) d\vec{q} - U \right\} \end{aligned}$$

where

$$\vec{q} := (\vec{x}, \vec{v}) \in Q$$

and $\alpha, \beta \in \mathbb{R}$ are Lagrange multipliers. To compute the variation observe that

$$\begin{aligned} \frac{d}{ds} [J(f+sg)] \Big|_{s=0} \\ &= \iint_Q g(\vec{q}) \left\{ k_B \log \left(\frac{f(\vec{q})}{NP_0} \right) + k_B + \alpha + \beta u(\vec{q}) \right\} d\vec{q} \\ &= 0 \end{aligned}$$

for any g with $f+sg: Q \rightarrow [0, \infty)$, et cetera. Thus,

$$k_B \log \left(\frac{f(\vec{q})}{NP_0} \right) + k_B + \alpha + \beta u(\vec{q}) = 0, \quad \forall \vec{q} \in Q.$$

Solving for f , we find

$$f^{\text{eq}}(\vec{q}) = P_0 \frac{N}{Z} \exp \left(\frac{-\beta u(\vec{q})}{k_B} \right),$$

where

$$Z = P_0 \iint_Q \exp \left(\frac{-\beta u(\vec{q})}{k_B} \right) d\vec{q}$$

is the unitless partition function. Here we have used the constraint (11.8) to eliminate α . Thus,

$$\iint_Q f^{\text{eq}}(\vec{q}) d\vec{q} = N.$$

We contend that, as before, the energy constraint (11.9) is sufficient to eliminate β , and in fact β is a unique positive function of V and N .

We will demonstrate this momentarily. //

Theorem (11.2): At equilibrium, the Gibbs entropy satisfies

$$(11.12) \quad S_{\text{eq}}(V, N) := S_G[f^{\text{eq}}] = \beta U + k_B \log(Z) N,$$

and at equilibrium the derivatives are

$$(11.13) \quad \frac{\partial S_{\text{eq}}}{\partial U} = \beta \quad \text{and} \quad \frac{\partial S_{\text{eq}}}{\partial N} = k_B \log(Z).$$

Proof: At equilibrium,

$$f^{\text{eq}}(\vec{q}) = P_0 \frac{N}{Z} \exp\left(\frac{-\beta U(\vec{q})}{k_B}\right)$$

where

$$Z = P_0 \iint_Q \exp\left(\frac{-\beta U(\vec{q})}{k_B}\right) d\vec{q}$$

$$S_{\text{eq}}(V, N) = S_G[f^{\text{eq}}]$$

$$= -k_B \iint_Q f^{\text{eq}}(\vec{x}, \vec{v}) \log\left(\frac{f^{\text{eq}}(\vec{x}, \vec{v})}{P_0 N}\right) d\vec{v} d\vec{x}$$

$$\begin{aligned}
&= -k_B \iint_Q \frac{P_0 N}{Z} \exp\left(\frac{-\beta u(\vec{q})}{k_B}\right) \log\left(\frac{\exp\left(\frac{-\beta u(\vec{q})}{k_B}\right)}{Z}\right) d\vec{q} \\
&= -\frac{k_B P_0 N}{Z} \iint_Q \exp\left(\frac{-\beta u(\vec{q})}{k_B}\right) \left\{ \frac{-\beta u(\vec{q})}{k_B} - \log(Z) \right\} d\vec{q} \\
&= \frac{P_0 N}{Z} \beta \iint_Q \exp\left(\frac{-\beta u(\vec{q})}{k_B}\right) u(\vec{q}) d\vec{q} \\
&\quad + \frac{k_B P_0 N}{Z} \log(Z) \iint_Q \exp\left(\frac{-\beta u(\vec{q})}{k_B}\right) d\vec{q} \\
&= \beta U + k_B N \log(Z).
\end{aligned}$$

Suppose that τ is a process parameter and assume that $(\vec{q}, \tau) \mapsto f(\vec{q}, \tau)$,

$$f : Q \times [0, \infty) \rightarrow [0, \infty).$$

Thus,

$$N(\tau) = \iint_Q f(\vec{q}, \tau) d\vec{q}.$$

It now follows that

$$\begin{aligned}
&\frac{d}{d\tau} S_G[f(\cdot, \tau)] \\
&= -k_B \frac{d}{d\tau} \iint_Q f(\vec{q}, \tau) \log\left(\frac{f(\vec{q}, \tau)}{P_0 N(\tau)}\right) d\vec{q}
\end{aligned}$$

$$= -k_B \iint_Q \left\{ \dot{f} \log \left(\frac{f}{P_0 N} \right) + f \left[\frac{\dot{f}}{f} - \frac{\dot{N}}{N} \right] \right\} d\vec{q}$$

$$= -k_B \iint_Q \dot{f} \left\{ \log \left(\frac{f}{P_0 N} \right) + 1 \right\} d\vec{q}$$

$$+ k_B \frac{\dot{N}}{N} \iint_Q f d\vec{q}$$

$$= -k_B \iint_Q \dot{f} \log \left(\frac{f}{P_0 N} \right) d\vec{q}.$$

Here, we have used the notation

$$\dot{f} = \frac{\partial f}{\partial \tau}, \quad \dot{N} = \frac{dN}{d\tau}.$$

Next, observe that

$$U(\tau) = \iint_Q f(\vec{q}, \tau) u(\vec{q}) d\vec{q},$$

so that

$$\frac{dU}{d\tau} = \dot{U} = \iint_Q \dot{f} u d\vec{q}.$$

Suppose that $(\vec{q}, \tau) \mapsto f(\vec{q}, \tau)$,

$$f: Q \times [0, \infty) \rightarrow [0, \infty),$$

is a path through number density space, with

$$f(\cdot, r_0) = f^{eq}(\cdot).$$

Then,

$$\text{and } N(r_0) = N, \quad U(r_0) = U,$$

$$\frac{d}{dr} S_G[f(\cdot, r_0)]$$

$$= -k_B \iint_Q \dot{f}(q, r_0) \log \left(\frac{f(\vec{q}, r_0)}{P_0 N(r_0)} \right) d\vec{q}$$

$$= -k_B \iint_Q \dot{f}(q, r_0) \log \left(\frac{f^{eq}(\vec{q})}{P_0 N} \right) d\vec{q}$$

$$= -k_B \iint_Q \dot{f} \left[\left(-\frac{\beta u(\vec{q})}{k_B} \right) - \log(Z) \right] d\vec{q}$$

$$= \beta \dot{U} + k_B \dot{N} \log(Z).$$

As in the proof of Theorem (10.2), we can conclude that

$$\frac{\partial S_{eq}}{\partial U} = \beta \quad \text{and} \quad \frac{\partial S_{eq}}{\partial N} = k_B \log(Z). //$$

Definition (11.3): The statistical temperature is defined by the expression

(11.14)

$$\frac{1}{T} := \frac{\partial S_{eq}}{\partial U},$$

where

$$S_{\text{eq}}(U, N) = S_g[f^*].$$

The statistical chemical potential is defined via

$$(11.15) \quad \frac{\mu}{T} := \frac{\partial S_{\text{eq}}}{\partial N}.$$

Thus

$$(11.16) \quad T = \frac{1}{\beta}$$

and

$$(11.17) \quad \mu = k_B T \log(Z).$$

As with the discrete case,

$$T = T(U, N)$$

and

$$\mu = \mu(U, N).$$

One can prove that the temperature relation is invertible, in the sense that there is a unique function

$$(11.18) \quad \tilde{U} = \tilde{U}(T, N)$$

with the property

$$\tilde{U}(T(U, N), N) = U, \quad \forall (U, N) \in \text{dom}(T).$$